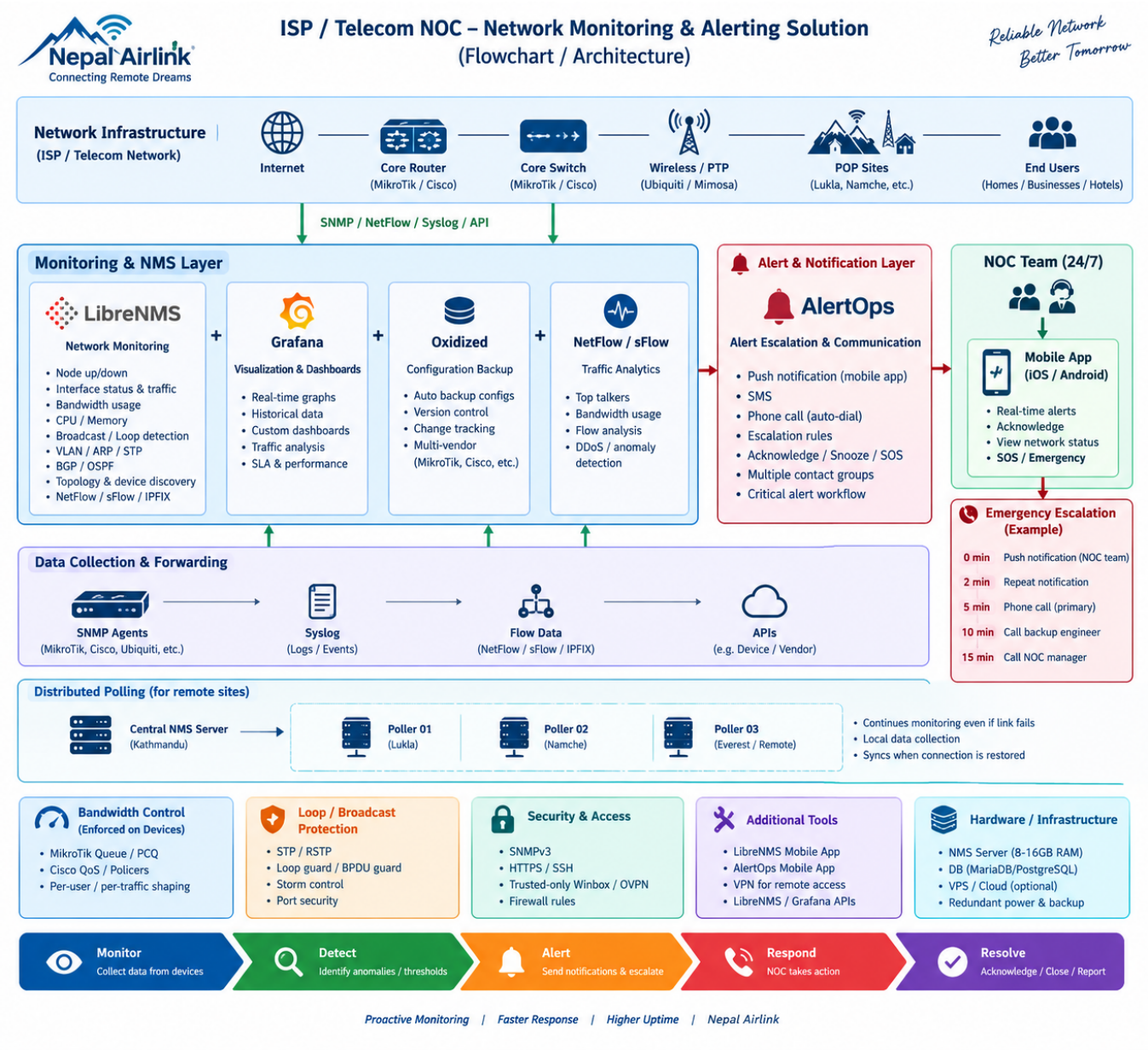
NEPAL AIRLINK

ISP / Telecom NOC — Network Monitoring, Alerting & Emergency Escalation Solution



Purpose: Provide a practical, ISP-grade monitoring architecture that continuously observes network availability, bandwidth, device health, routing, wireless/backhaul status, broadcast/loop indicators and critical failures, then escalates incidents to the NOC team through mobile push, SMS and phone-call workflows.

Core concept: Monitor → Detect → Alert → Escalate → Respond → Resolve → Report

1. Executive Summary

The proposed solution is designed for a regional ISP/telecom environment with a mixture of MikroTik, Cisco, Ubiquiti/Mimosa and other network equipment across central and remote POPs. The architecture separates four jobs: network devices provide telemetry and enforce network controls; the NMS collects and analyzes operational data; visualization and flow tools provide deep visibility; and the alerting platform handles human escalation.

The recommended stack is LibreNMS as the primary network monitoring platform, Grafana for advanced dashboards and historical visualization, NetFlow/sFlow/IPFIX for traffic analytics, Oxidized for automated configuration backup, and AlertOps (or an equivalent incident-escalation service) for push/SMS/voice escalation. MikroTik/Cisco QoS and queueing remain responsible for actual bandwidth enforcement, while STP/RSTP, storm-control and loop-protection features remain responsible for preventing network loops.

2. Architecture at a Glance

Layer	Main components	Primary responsibility
1. Network	MikroTik, Cisco, Ubiquiti/Mimosa, OLTs, wireless platforms, servers	Carry customer and backbone traffic; enforce QoS and protection
2. Data collection	SNMPv3, Syslog, NetFlow/sFlow/IPFIX, APIs	Send health, interface, event and traffic telemetry
3. NMS	LibreNMS	Availability, interfaces, health, thresholds, topology and alerts
4. Visualization	Grafana	Real-time/historical dashboards, SLA and capacity views
5. Traffic analytics	NetFlow/sFlow/IPFIX collector	Top talkers, protocols, traffic patterns and anomalies
6. Config management	Oxidized	Automated device configuration backup and change history
7. Incident escalation	AlertOps / equivalent	Push → SMS → phone → escalation → acknowledgement
8. NOC operations	iOS/Android + web dashboard	Acknowledge, investigate, coordinate and resolve incidents

3. Detailed End-to-End Flow

Step 1 — Network generates telemetry

Every router, switch, wireless device, OLT and supported server exposes operational information. SNMP provides counters and health information; Syslog sends events; flow protocols provide traffic records; APIs can expose vendor-specific data.

Step 2 — Data is collected

Central or distributed pollers query devices at defined intervals. Remote pollers are useful at Lukla, Namche and other isolated POPs because they reduce latency and continue local collection during WAN interruptions.

Step 3 — LibreNMS evaluates the network

LibreNMS determines whether devices and interfaces are reachable, compares measurements against thresholds, tracks traffic, records errors and discovers topology-related information.

Step 4 — Traffic is analyzed

NetFlow/sFlow/IPFIX identifies traffic sources, destinations, applications/protocols and top talkers. This helps distinguish normal utilization from congestion, unusual traffic or possible abuse.

Step 5 — Dashboards provide context

Grafana can combine NMS and flow data into NOC dashboards showing core links, POP status, bandwidth, CPU/RAM, customer-impact indicators, SLA trends and historical capacity.

Step 6 — An incident is classified

Alerts are assigned a severity such as Critical/P1, Major/P2 or Warning/P3. Alert dependencies and correlation should be used so one failed core device does not generate hundreds of confusing downstream alerts.

Step 7 — AlertOps escalates

Critical incidents are sent to the appropriate contact group. The escalation policy can move from mobile push to repeated notification, SMS and automated phone call if nobody acknowledges the incident.

Step 8 — NOC responds

The engineer opens the alert, identifies the affected POP/interface/device, checks recent changes and traffic, and performs corrective action through approved management channels such as VPN, SSH or Winbox.

Step 9 — Resolve and document

Once the network is stable, the alert is acknowledged/closed, the incident is documented and recurring problems can be reviewed for preventive action.

4. What Should Be Monitored

Category	Examples	Suggested trigger
Availability	Device down, interface down, POP unreachable	Immediate for core; delayed for access devices to avoid f
Bandwidth	WAN, core, backhaul and uplink utilization	Warning 80%; Major 90%; Critical 95% sustained
Device health	CPU, memory, temperature, fan/PSU where available	CPU/RAM >90% sustained; vendor-specific thermal limits
Interface quality	CRC, errors, discards, packet drops	Rate-of-change and sustained threshold rather than a sin
Broadcast/loop	Broadcast rate, unknown unicast, STP events, MAC movement	Rapid increase or repeated topology/MAC-flap events
Routing	BGP/OSPF adjacency, route changes	Session down/flapping; unexpected route change
Wireless	Signal, CCQ/SNR where supported, retries, capacity	Vendor/model-specific thresholds
Traffic behavior	Top talkers, protocols, unusual volume	Deviation from normal baseline

Important: Thresholds are starting values, not universal rules. They should be tuned using actual traffic patterns, device capabilities, interface speeds and the difference between core, aggregation and access layers.

5. Alert Severity & Emergency Escalation

Severity	Typical events	Notification	Example response
P1 Critical	Core/major POP down; major backhaul failure; BGP core failure; power loss	Push + SMS + phone call escalation	Immediate NOC response; identify customer
P2 Major	Aggregation link down; high utilization; repeated BGP flap; device unreachable	Push, repeat, SMS if not acknowledged	Investigate within defined NOC SLA
P3 Warning	High utilization trend; elevated CPU/RAM; temperature warning; CRC errors	Push/email/dashboard	Planned corrective action and capacity revie

Recommended P1 escalation example

Time	Action
0 min	Mobile push notification to primary NOC group
2 min	Repeat notification if unacknowledged
5 min	Automated phone call to primary engineer
10 min	Call backup engineer / second contact group
15 min	Escalate to NOC supervisor/manager

Time	Action
ACK	Stop escalation; incident remains open until resolved

6. Monitoring vs. Network Control

A key design principle is to keep monitoring separate from enforcement. The NMS should tell the NOC what is happening and why it matters; the routers and switches should enforce the actual network policy.

Function	Responsible system	Example
Monitor bandwidth	LibreNMS + flow collector	Show 820 Mbps / 1 Gbps and raise alert at configured threshold
Limit bandwidth	MikroTik / Cisco QoS	Queue, PCQ, shaping, policing, per-user or per-service policy
Detect possible loop	NMS + switch telemetry	Broadcast spike, STP change, MAC movement
Prevent loop/storm	Switch/router	RSTP/STP, loop guard, BPDU guard, storm control, port security

7. Distributed Monitoring for Remote POPs

For geographically difficult locations, a central NMS alone can create blind spots when the path between the remote POP and Kathmandu fails. A distributed-poller design places a monitoring collector closer to the remote equipment. The poller continues to observe local devices and synchronizes with the central NMS when connectivity returns.

Example: Central NMS in Kathmandu → Poller 01 in Lukla → local routers/switches/APs; Poller 02 in Namche → local aggregation/access devices; Poller 03 for another remote/critical area.

8. Recommended NOC Dashboards

- Executive/Network Overview: total devices, devices down, critical alarms, core link utilization and overall POP status.
- Core Dashboard: WAN links, core routers/switches, BGP/OSPF, CPU/RAM, packet loss and interface errors.
- POP Dashboard: Lukla/Namche/other POP status, backhaul utilization, radio health and downstream device count.
- Capacity Dashboard: 24-hour/7-day/30-day bandwidth trends, peak utilization and links approaching capacity.
- Incident Dashboard: P1/P2/P3 open alerts, unacknowledged incidents, acknowledgement time and resolution time.
- Traffic Dashboard: top talkers, top protocols, unusual traffic and flow-volume trends.
- Configuration Dashboard: last successful backup, devices with changed configurations and backup failures.

9. Security & Access

- Use SNMPv3 wherever supported; avoid exposing SNMP directly to the public Internet.
- Place NMS/pollers in protected management VLANs or management VRFs.
- Use VPN/zero-trust style access for remote NOC management; restrict Winbox/SSH/HTTPS to trusted sources.
- Apply firewall rules that allow monitoring traffic only from authorized pollers.
- Use role-based access for NMS and alerting platforms.
- Keep configuration backups encrypted and access-controlled.
- Record administrative actions and configuration changes for incident investigation.

10. Suggested Initial Infrastructure

Component	Initial recommendation	Notes
Central NMS server	8–16 GB RAM; SSD; multi-core CPU	Scale based on device/interface count and polling frequency
Database	MariaDB/PostgreSQL on fast SSD	Use regular automated backups
Remote poller	2–4 CPU cores; 4–8 GB RAM	Size for number of local devices and flow volume
Flow collector	SSD + additional CPU/RAM as traffic grows	Flow volume can become much larger than SNMP data
Backup storage	Separate storage/NAS/cloud backup	Keep NMS DB and configuration backups separate
Network access	Management VLAN/VRF + VPN	Do not expose management services unnecessarily

11. Recommended Implementation Plan

Phase 1 — Foundation

- Deploy central LibreNMS server and database.
- Create management VLAN/VRF and secure SNMPv3 access.
- Add core routers, core switches and major POP devices.
- Configure device naming, location, role and contact metadata.

Phase 2 — Visibility

- Add interface traffic, CPU/RAM, errors/discards, temperature and routing monitoring.
- Deploy Grafana dashboards for NOC and management.
- Enable Syslog collection and basic event correlation.
- Establish baseline traffic and device-health values.

Phase 3 — Traffic & Configuration

- Enable NetFlow/sFlow/IPFIX on appropriate core/aggregation devices.
- Deploy Oxidized for automated configuration backup.
- Build top-talker and capacity dashboards.
- Set configuration-change review procedures.

Phase 4 — Remote Monitoring

- Deploy distributed pollers at critical remote POPs.
- Prioritize locations where a WAN failure would otherwise create a monitoring blind spot.
- Test local data collection and synchronization after link recovery.

Phase 5 — Critical Alerting

- Integrate AlertOps or equivalent escalation platform.
- Define P1/P2/P3 rules and contact groups.
- Test push, SMS, phone-call and acknowledgement workflows.
- Run controlled failure drills.

Phase 6 — Optimization

- Tune thresholds using real network behavior.
- Add alert dependencies to reduce duplicate alarms.
- Create monthly capacity, uptime and incident reports.
- Review repeated incidents and implement preventive network changes.

12. Example Critical Incident

Scenario: A core aggregation router at Namche becomes unreachable. At the same time, several downstream devices begin reporting unavailable status.

Stage	System action	NOC outcome
Detection	LibreNMS loses contact with the Namche core router and confirms the condition using multiple polling cycles.	Incident is created as P1 if dependency rules indicate m
Correlation	Dependent downstream alarms are grouped or suppressed where appropriate.	NOC sees one meaningful incident instead of dozens of dup
Notification	AlertOps sends push notification to the primary NOC group.	Engineer receives the incident on mobile.

Stage	System action	NOC outcome
Escalation	No acknowledgement → repeat notification → SMS/voice escalation.	Backup engineer is contacted automatically.
Investigation	Engineer checks core router, uplink, power, optical/radio path and recent configuration changes.	Root cause is narrowed down quickly.
Recovery	Network path is restored or backup path is activated.	Devices return online and customer service recovers.
Closure	Alert is acknowledged and incident record is closed with cause/action notes.	Event becomes useful operational history.

13. Final Recommended Stack

Role	Recommended platform/technology
Primary NMS	LibreNMS
Advanced visualization	Grafana
Traffic analytics	NetFlow / sFlow / IPFIX collector
Configuration backup	Oxidized
Incident escalation	AlertOps or equivalent push/SMS/voice platform
Device bandwidth enforcement	MikroTik Queue/PCQ and Cisco QoS/policers
Loop/storm protection	RSTP/STP, loop guard/BPDU guard, storm control, port security
Secure management	SNMPv3 + VPN + restricted SSH/Winbox/HTTPS
Remote monitoring	Distributed pollers at critical POPs

Target operating model: The NMS continuously watches the network; flow analytics explains traffic; configuration management preserves device state; the alerting layer makes sure critical incidents reach a human; and the NOC team investigates and resolves the underlying network problem.

This architecture is intentionally modular, so Nepal Airlink can start with core monitoring and expand into remote pollers, flow analytics, automated configuration backup, advanced alert correlation and 24/7 emergency escalation.